On the Kelly Monoidal Structure of Λ-Sequences

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Operad

Definition (May)

An operad in Top consists of a sequence of spaces $\mathcal{C}(n)$ with Σ_n-action for $n \geq 0$, and structure maps

$$\gamma : \mathcal{C}(k) \times \mathcal{C}(n_1) \times \cdots \times \mathcal{C}(n_k) \to \mathcal{C}(n_1 + \cdots + n_k)$$
$$\eta : \mathrm{pt} \to \mathcal{C}(1)$$

such that compositional, unital and permutation equivariant relations hold.

Operad

Operads provide a solid framework for us to consider algebraic structures.

1. $\mathcal{M} : \mathcal{M}(k) = \Sigma_k$. $\mathcal{M}$-spaces = topological monoids.
2. $\mathcal{N} : \mathcal{N}(k) = \mathrm{pt}$. $\mathcal{N}$-spaces = commutative topological monoids.

The definition for operads in $\mathbf{Top}$ can be generalized to any symmetric monoidal category $(\mathcal{C}, \otimes, \mathcal{I})$, which is the case we are interested in.

Kelly's product

$$\Sigma: \quad \begin{array}{lcl} \text{obj} & : & \mathbf{n} = \{0, 1, \cdots, n\}, n \geq 0 \\ \text{mor} & : & \text{based bijection} \end{array}$$

$$\Lambda: \quad \begin{array}{lcl} \text{obj} & : & \mathbf{n} = \{0, 1, \cdots, n\}, n \geq 0 \\ \text{mor} & : & \text{based injection} \end{array}$$

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- Σ-sequence $\Sigma^{op}[\mathcal{V}]$ $\mathcal{C} : \Sigma^{op} \to \mathcal{V}$
- Λ-sequence $\Lambda^{op}[\mathcal{V}]_{\mathcal{I}_0}$ $\mathcal{C} : \Lambda^{op} \to \mathcal{V}$ with $I \to \mathcal{C}(0)$
- unital sequence $\Lambda_I^{op}[\mathcal{V}]$ Λ-sequence with $\mathcal{C}(0) = I$

Kelly's product

Kelly's product realize the world of operads to the world of monoids. It is defined in two steps.

1. "Box product " $\boxtimes$ is defined to be the Day convolution

$$\begin{array}{ccccc} \Lambda^{op} \times \Lambda^{op} & \xrightarrow{\mathcal{C}\times\mathcal{D}} & \mathcal{V} \times \mathcal{V} & \xrightarrow{\otimes} & \mathcal{V} \\ \vee \downarrow & & \text{LKan} & \nearrow_{\mathcal{C}\boxtimes\mathcal{D}} & \\ \Lambda^{op} & & & & \end{array}$$

$$D \boxtimes E = \int^{n_1,n_2} \Lambda^{op}(n_1+n_2,-) \otimes \big(D(n_1) \otimes E(n_2)\big)$$

2. "Composition product" $\odot_\Lambda$ is defined to be

$$D \odot_\Lambda E = \int^{m} D(m) \otimes E^{\boxtimes m}$$

Kelly's product

$\odot_\Lambda$ gives $\Lambda^{op}[\mathcal{V}]$ a monoidal structure, with the unit $\mathcal{I}_0$

$$\mathcal{I}_0(n) = \begin{cases} I & n = 0 \\ \varnothing & n > 0 \end{cases}$$

Theorem

1. *(Kelly) operads = monoids in* $\Sigma^{op}[\mathcal{V}]$
2. *(May-Zhang-Zou) unital operads = monoids in* $\Lambda_I^{op}[\mathcal{V}]$

What about the case where the symmetric monoidal structure on $\mathcal{V}$ is no more closed? i.e. $\mathcal{V}$ does not have an internal hom $\underline{\mathrm{Hom}}(A, -)$ which is a right adjoint of $- \otimes A$.

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Main difficulties:

1. In $\mathcal{V}$ colimits do not commute with tensors.
2. Box and composition products are no more associative.
3. We can no more define n-fold box and composition products on $\Lambda^{op}[\mathcal{V}]$.

Normal op lax monoidal structure

Definition (Ching)

A normal oplax monoidal structure on a category $\mathcal{C}$ consists with the following data:

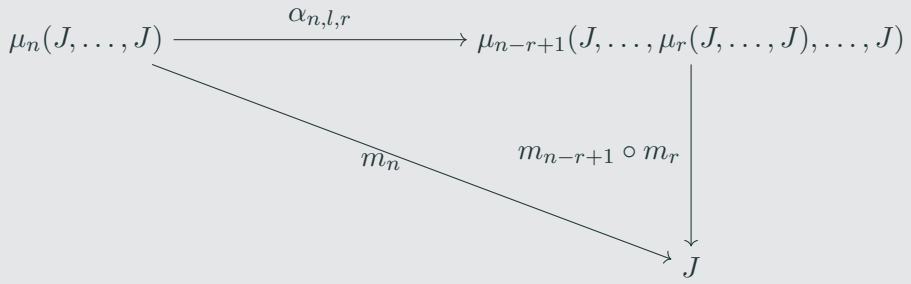
1. A unit object $\mathcal{I}$ in $\mathcal{C}$.
2. A series of functors $\mu_0, \mu_1, \mu_2, \ldots$, where $\mu_n : \mathcal{C}^n \to \mathcal{C}$. It satisfies μ_1 is the identity functor on $\mathcal{C}$ and μ_0 is the constant functor whose value is $\mathcal{I}$.
3. A series of natural transformations.
   $\alpha_{n,l,r} : \mu_n(X_1, \ldots, X_n) \to$
   $\mu_{n-r+1}(X_1, \ldots, X_l, \mu_r(X_{l+1}, \ldots, X_{l+r}), X_{l+r+1}, \ldots, X_n)$ for triples (n, l, r) satisfies $0 \leqslant l \leqslant l + r \leqslant n$.

These data satisfies op lax associativity and unit laws.

Normal op lax monoidal structure

Definition (Ching)

A monoid is an object J with morphisms $m_n : \mu_n(J, J, \ldots, J) \to J$ such that the following diagram commutes for all (n, l, r) satisfying $0 \leqslant l \leqslant l + r \leqslant n$.



Obviously, when $\mathcal{C}$ is a monoidal category, the definition consists with the old one.

Theorem (F.-Zou)

Kelly's composition product exists on $\Lambda^{op}[\mathcal{V}]$ when $\mathcal{V}$ is no more closed, endowing $\Lambda^{op}[\mathcal{V}]_{I_0}$ and $\Lambda^{op}_I[\mathcal{V}]$ a structure of normal op lax monoidal structure.

Theorem (F.-Zou)

Kelly's composition product exists on $\Lambda^{op}[\mathcal{V}]$ when $\mathcal{V}$ is no more closed, endowing $\Lambda^{op}[\mathcal{V}]_{I_0}$ and $\Lambda^{op}_I[\mathcal{V}]$ a structure of normal op lax monoidal structure.

Not unique, but instead satisfies a universal property.

Examples

When $n = 3$

$$T \odot_\Lambda S \odot_\Lambda R = \int^{m,k,n_1\ldots,n_m}_{j_1,\ldots,j_k} \Lambda^{op}(\mathbf{j_1} + \cdots + \mathbf{j_k}, -) \otimes \Lambda^{op}(\mathbf{n_1} + \cdots + \mathbf{n_m}, \mathbf{k}) \otimes \Big(T(\mathbf{m}) \otimes S(\mathbf{n_1}) \otimes \cdots \otimes S(\mathbf{n_m}) \otimes R(\mathbf{j_1}) \otimes \cdots \otimes R(\mathbf{j_k}) \Big)$$

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There are arrows $T \odot_\Lambda S \odot_\Lambda R \to (T \odot_\Lambda S) \odot_\Lambda R$ and $T \odot_\Lambda S \odot_\Lambda R \to T \odot_\Lambda (S \odot_\Lambda R)$, commuting "coends" with tensor. They are $\alpha_{3,0,2}$ and $\alpha_{3,1,2}$.

When $\mathcal{V}$ is closed, these morphisms are isomorphisms.

Monoidal localization

Main ingredient: an embedding L from $\mathcal{V}$ to a bicomplete symmetric monoidal category $(\overline{\mathcal{V}}, \otimes, \mathcal{I})$. Colimits in $\overline{\mathcal{V}}$ commute with tensor.

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We consider a collection of morphisms S in $(\mathcal{V}, \otimes, \mathcal{I})$. Property of S determines property of L and $\mathcal{V}[S^{-1}]$.

1. L preserves limits and colimits, $\mathcal{V}[S^{-1}]$ is bicomplete $\Longleftarrow$ S is a multiplicative system admits products.
2. L and $\mathcal{V}[S^{-1}]$ are symmetric monoidal $\Longleftarrow$ S is closed under tensor.

Monoidal localization

We take S to be $S = \{\mathrm{colim}_{i\in I}(A_i \otimes X) \to \mathrm{colim}_{i\in I}(A_i) \otimes X\}$ and extend it to a collection $\overline{S}$ satisfies above conditions. Take $L : \mathcal{V} \to \mathcal{V}[S^{-1}]$ to be the localization.

Two fold product $\odot_\Lambda$ on $\Lambda_I^{op}[L\mathcal{V}]$ is monoidal, and

$$L(\mathcal{D} \odot \mathcal{E}) \cong L\mathcal{D} \odot L\mathcal{E}$$

We inductively define $\mu_n : (\Lambda_I^{op}[\mathcal{V}])^n \to \Lambda_I^{op}[\mathcal{V}]$ and $\alpha_{n,l,r}$ lifting those on $\Lambda_I^{op}[L\mathcal{V}]$. (Note: $L\alpha_{n,l,r}$'s are isomorphisms.)

Relationship with operads

Theorem

When $\mathcal{V}$ is not closed.

1. *(Ching) operads = monoids in $\Sigma^{op}[\mathcal{V}]$*
2. *(F.-Zou) unital operads = monoids in $\Lambda_I^{op}[\mathcal{V}]$*

Remark

A monoid on a normal op lax monoidal category depends only on the information of μ_n and $\alpha_{n,l,r}$ when $n \leqslant 3$.

Thanks for your attention

Please feel free to ask me anything!